AI Relief Command Center

Hackathon Project Submission | Crisis Management, HealthTech & Emergency Response

Theme

Crisis Management, HealthTech & Emergency Response

This theme focuses on building intelligent solutions that improve disaster preparedness, emergency response, and coordination between citizens, rescue teams, and authorities to save lives during emergencies.

> *Annotation: Framing the theme up front tells judges immediately which track/category the project targets — useful for aligning expectations before the problem statement.*

Problem Statement

During disasters such as floods, earthquakes, cyclones, fires, or landslides, emergency responders receive hundreds or even thousands of requests through calls, messages, images, and voice recordings.

Major challenges include:

- Manually analyzing every emergency report is slow.
- Rescue teams cannot easily identify which incidents require immediate attention.
- Resources such as ambulances, rescue teams, food, and medical supplies are often deployed inefficiently.
- Multiple agencies work independently, leading to delays and poor coordination.
- Authorities lack a centralized platform to monitor incidents, assign rescue teams, and track ongoing operations in real time.

These issues increase response time, reduce operational efficiency, and can ultimately lead to preventable loss of life.

> *Annotation: Each bullet represents a distinct pain point — keep these as separate, falsifiable claims. Judges often map each bullet to a specific feature later, so the 1:1 correspondence with 'Key Features' below matters.*

Our Approach

We developed **AI Relief Command Center**, an AI-powered emergency management platform that transforms raw emergency reports into actionable intelligence for disaster response teams.

The system enables citizens to submit incidents using text, images, or audio. These reports are analyzed using Artificial Intelligence to determine the type of emergency, assess its severity, estimate required resources, and recommend the most appropriate rescue response.

The platform provides a centralized dashboard where emergency officials can monitor all active incidents, assign rescue teams, track mission progress, and analyze live operational data. Interactive maps display incident locations, while analytics and simulations help authorities prepare for large-scale disasters and optimize resource allocation.

By integrating AI, cloud-based data management, and real-time monitoring into a single platform, AI Relief Command Center significantly reduces emergency response time, improves decision-making, and enhances coordination among rescue teams.

> *Annotation: This section follows an Input → AI Processing → Centralized Action loop. Consider a one-line architecture diagram on the demo slide to reinforce this flow visually.*

Key Features

- AI-powered incident analysis using the Gemini API
- Multi-user authentication with role-based access (Citizen, Rescue Team, Operations Officer, Admin)
- Incident reporting with text, image, and audio uploads
- Automatic emergency categorization and priority detection
- Interactive disaster map using Leaflet and OpenStreetMap
- Rescue team assignment and mission tracking
- Live dashboard with analytics and operational insights
- Simulation mode for disaster preparedness and training
- Notification center for real-time mission updates
- Cloud-based PostgreSQL database hosted on Supabase
- Production-ready deployment using FastAPI, React, Render, and Vercel

> *Annotation: Group these when presenting — (1) Intake: text/image/audio upload, (2) AI Layer: Gemini categorization & priority detection, (3) Ops Layer: dashboard, map, team assignment, (4) Infra: auth, DB, deployment. Grouping helps judges follow the stack quickly during a timed demo.*

Impact

AI Relief Command Center helps emergency responders make faster and more informed decisions by reducing manual effort and improving coordination.

Expected benefits include:

- Faster emergency response times
- Better prioritization of life-threatening incidents

- Efficient utilization of rescue teams and resources
- Improved communication between citizens and authorities
- Real-time operational visibility
- Enhanced disaster preparedness through simulations
- Scalable cloud-based architecture suitable for district, state, or national deployment

> *Annotation: 'Scalable ... district, state, or national deployment' is your growth/impact story — worth emphasizing if the hackathon rewards real-world scalability and social impact.*

Technology Stack

Component	Technology
Frontend	React + TypeScript + Vite + Tailwind CSS
Backend	FastAPI
Database	Supabase PostgreSQL with SQLAlchemy
AI	Gemini API
Maps	Leaflet + OpenStreetMap
Charts	Recharts
Authentication	JWT + BCrypt
Deployment	Vercel (Frontend), Render (Backend)

> *Annotation: Full-stack, cloud-native choice (Vercel + Render + Supabase) — good for a hackathon because it's free-tier friendly and demo-ready with minimal DevOps overhead.*

This document is suitable for a hackathon submission, project report, or explaining the project to judges in a professional and concise manner.